GEOMETRIC REGULARITY OF A COMPLETED LOCALISATION OF A POWER SERIES RING OVER A POLYNOMIAL RING

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The completion of a regular local ring is regular. Lemma \ref{lemma-completion-regular}, cited at proof line 12317. Gives the base case of the induction.
- Completion of a finite extension of Noetherian complete local rings is a product of completions. Algebra, Lemma \ref{algebra-lemma-completion-finite-extension}, cited at proof lines 12332-12333. Identifies the base change with the completed localisation of the finite algebra realising the field extension.
- Existence of a derivation of a finite algebra that does not vanish on a prescribed element. Lemma \ref{lemma-find-D}, cited at proof line 12341. Produces a derivation D of B with D(f) nonzero, so that D(f) becomes a unit after inverting denominators.
- A derivation extends to a localisation and to a completion. Lemma \ref{lemma-derivation-extends}, cited at proof lines 12343-12346. Extends D to the completed localisation tensored with the intermediate field, where D(f) is a unit.
- A degree-p extension obtained by adjoining a p-th root is regular when a derivation sends that root's argument to a unit. Lemma \ref{lemma-degree-p-extension-regular}, cited at proof lines 12350-12351. Concludes regularity of the base change after adjoining the p-th root.

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Source reference. Original lemma 07PR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9806–9806. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07NY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9778–9778. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07N9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9754–9754. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07PH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9797–9797. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07PE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9794–9794. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07PG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9796–9796. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let k be a field of characteristic p. Let $A = k[[x_1, \ldots, x_n]][y_1, \ldots, y_m]$ and denote K the fraction field of A. Let $\mathfrak{p} \subset A$ be a prime. Then $A^\wedge_\mathfrak{p} \otimes_A K$ is geometrically regular over K.*

Proof. Let L/K be a finite purely inseparable field extension. We will show by induction on $[L : K]$ that $A^\wedge_\mathfrak{p} \otimes L$ is regular. The base case is $L = K$: as A is regular, $A^\wedge_\mathfrak{p}$ is regular (Lemma 07NY), hence the localization $A^\wedge_\mathfrak{p} \otimes K$ is regular. Let $K \subset M \subset L$ be a subfield such that L is a degree p extension of M obtained by adjoining a pth root of an element $f \in M$. Let B be a finite A-subalgebra of M with fraction field M. Clearing denominators, we may and do assume $f \in B$. Set $C = B[z]/(z^p - f)$ and note that $B \subset C$ is finite and that the fraction field of C is L. Since $A \subset B \subset C$ are finite and $L/M/K$ are purely inseparable we see that for every element of B or C some power of it lies in A. Hence there is a unique prime $\mathfrak{r} \subset B$, resp. $\mathfrak{q} \subset C$ lying over $\mathfrak{p}$. Note that

$$A^\wedge_\mathfrak{p} \otimes_A M = B^\wedge_\mathfrak{r} \otimes_B M$$

see Algebra, Lemma 07N9. By induction we know that this ring is regular. In the same manner we have

$$A^\wedge_\mathfrak{p} \otimes_A L = C^\wedge_\mathfrak{r} \otimes_C L = B^\wedge_\mathfrak{r} \otimes_B M[z]/(z^p - f)$$

the last equality because the completion of $C = B[z]/(z^p - f)$ equals $B^\wedge_\mathfrak{r}[z]/(z^p - f)$. By Lemma 07PH we know there exists a derivation $D : B \to B$ such that $D(f) \neq 0$. In other words, $g = D(f)$ is a unit in M! By Lemma 07PE D extends to a derivation of $B_\mathfrak{r}$, $B^\wedge_\mathfrak{r}$ and $B^\wedge_\mathfrak{r} \otimes_B M$ (successively extending through a localization, a completion, and a localization). Since it is an extension we end up with a derivation of $B^\wedge_\mathfrak{r} \otimes_B M$ which maps f to g and g is a unit of the ring $B^\wedge_\mathfrak{r} \otimes_B M$. Hence $A^\wedge_\mathfrak{p} \otimes_A L$ is regular by Lemma 07PG and we win. □